mmBeamKD: Parameter-Efficient Multimodal Beam Prediction via Cross-Modal Knowledge Distillation

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***Abstract*—Accurate beam prediction is a key enabler of low-latency millimeter-wave (mmWave) vehicular communications, yet training multimodal fusion models remains computationally expensive in scarce-data settings. We present mmBeamKD, a framework combining parameter-efficient multimodal fusion with cross-modal knowledge distillation for beam prediction on the DeepSense 6G Scenario-32 benchmark (1905 training samples, 64-beam classification). TRANSFUSERV5 freezes pretrained ResNet backbones and trains only a lightweight cross-attention fusion module (21.7 M trainable parameters vs. 78.4 M baseline), achieving a $1.66\times$ inference speedup and a 29% smaller deployment checkpoint.**

Modality ablation reveals that camera and GPS dominate in Scenario-32: removing either collapses DBA by more than 37%, while LiDAR and radar contribute negligibly. Building on this finding, we apply cross-modal knowledge distillation from a four-model heterogeneous teacher ensemble to a camera+GPS-only student (CAMGPS-ONLY-KD), which achieves DBA = 0.8356 ($+0.028$ vs. baseline; 95 % CI: $[+0.017, +0.040]$; $p < 0.0001$) at 19.9 ms inference, matching the full three-model ensemble while requiring only two sensors. A four-modality distilled student (TRANSFUSERV5-KD) achieves DBA = 0.8285 ($+0.021$; $p = 0.0002$). Per-beam analysis exposes a systematic accuracy gap for high-index beams (DBA = 0.694, beams 43–63), linked to a training-to-test beam-distribution shift. All comparisons are supported by paired bootstrap significance testing.

***Index Terms*—mmWave beam prediction, multimodal sensing, vehicular communications, knowledge distillation, cross-modal distillation, parameter-efficient learning, distribution shift**

I. INTRODUCTION

Millimeter-wave (mmWave) communication is a cornerstone of 5G/6G vehicular networks, offering multi-gigabit throughput for connected and autonomous vehicles. Because mmWave signals are highly directional, maintaining a reliable link requires selecting the correct transmit/receive beam from a codebook of tens to hundreds of candidate beams. In fast-moving vehicular scenarios, this beam management step must complete in milliseconds, making exhaustive beam sweeping impractical [1], [2].

Sensing-aided beam prediction offers a promising alternative: measurements from cameras, LiDAR, radar, or GPS taken at the user equipment can be used to predict the optimal beam index directly, without over-the-air scanning [3]–[5]. The recent DeepSense 6G dataset [5] provides a large-scale real-world benchmark for this paradigm, pairing synchronized camera, LiDAR, radar, and GPS measurements with ground-truth beam labels across 32 deployment scenarios.

Despite strong empirical results on this benchmark, existing approaches leave three practical gaps. First, they rely on large, fully trainable fusion networks that overfit when only a few thousand labeled samples are available, a constraint common to freshly deployed vehicular sites. Second, they rarely examine *which sensors actually drive performance*: a model trained with four modalities may rely primarily on one, making the additional hardware a wasted investment. Third, evaluations typically lack statistical rigor; performance differences between methods are reported without confidence intervals or significance tests, obscuring whether gains are reliable.

We emphasize upfront that *all quantitative results in this paper are specific to Scenario 32*; our work is a rigorous single-scenario case study rather than a claim of universal superiority. We include explicit out-of-scope warnings throughout and discuss multi-scenario generalization as future work. The value of the case study lies in (i) establishing the first statistically significant single-model DBA improvement on this benchmark, (ii) a controlled sensor-attribution analysis, and (iii) a methodology including bootstrap testing, distribution-shift analysis, and a beam-probing overhead metric, all of which transfer to other scenarios.

This paper presents **mmBeamKD**, which addresses all three gaps on DeepSense 6G Scenario 32, a single urban intersection site with 1905 training, 635 validation, and 635 test sequences. mmBeamKD comprises two stages: (i) TRANSFUSERV5, a parameter-efficient fusion backbone that replaces the baseline's eight-layer GPT transformer with a compact two-layer cross-attention module applied to frozen pretrained ResNet features, reducing trainable parameters from 78.4 M to 21.7 M and cutting inference latency by $1.66\times$; and (ii) cross-modal knowledge distillation from a heterogeneous four-model ensemble teacher, yielding both a four-modality student (TRANSFUSERV5-KD) and a camera+GPS-only student (CAMGPS-ONLY-KD) that matches the full ensemble at 19.9 ms. Modality ablation reveals an unexpected finding: in Scenario 32, *camera and GPS are nearly sufficient for full performance*, while LiDAR and radar contribute negligibly; cross-modal KD exploits this finding directly.

The contributions of this paper are as follows:

1) **Parameter-efficient fusion backbone.** TRANS-

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FUSERV5 achieves baseline-level DBA (0.8058 vs. 0.8076) with 3.6× fewer trainable parameters, 1.66× faster inference, and a 29% smaller checkpoint. This makes the model practical for edge deployment in resource-constrained vehicular roadside units.

2) **Knowledge distillation for single-model accuracy gains.** Distilling a heterogeneous four-model ensemble into a single TRANSFUSERV5 student (TRANSFUSERV5-KD) improves DBA by $+0.021$ over the baseline (95 % CI: $[+0.010, +0.032]$; $p = 0.0002$), representing a statistically significant and practically meaningful single-model gain on this benchmark; to the best of our knowledge the first such rigorously tested gain reported for Scenario 32.
3) **Camera+GPS modality dominance analysis.** Ablation experiments and a camera+GPS-only control model show that removing LiDAR or radar changes DBA by less than 0.001, while removing the camera drops DBA from 0.806 to 0.503 (-37.5%) and removing GPS drops it to 0.470 (-41.7%). These findings inform sensor selection for real deployments where LiDAR costs dominate.
4) **Distribution-shift failure analysis.** Per-beam-bin evaluation reveals that high-index beams (43–63, 8% of test samples) achieve only DBA = 0.694, while low-index beams achieve DBA = 0.819. We link this to a systematic shift between the validation and test beam distributions (mean 21.9 vs. 18.6), providing a concrete target for future shift-robust training methods.

All performance comparisons are accompanied by paired bootstrap 95 % confidence intervals and one-sided p-values, providing a statistical framework that is largely absent from prior work on this benchmark.

The rest of this paper is organized as follows. Section II reviews related work. Section III defines the system model and problem formulation. Section IV describes TRANSFUSERV5 and the distillation procedure. Section V presents the main experimental results. Section VI provides analysis of modality contributions, representation quality, and distribution shift. Section VII concludes.

II. RELATED WORK

a) mmWave beam management and beam prediction.: Traditional beam management relies on exhaustive beam sweeping, which introduces prohibitive overhead at high-frequency bands [2], [6]. Compressed sensing and codebook-based methods reduce sweep overhead but still require explicit channel measurements [1], [7]. Learning-based beam prediction [1], [8], [9] circumvents the sweep by directly mapping received signals or side information to a beam index. These approaches demonstrate that beam prediction can match or exceed the link quality of exhaustive scanning while operating at dramatically lower overhead, which is a prerequisite for vehicular latency targets.

b) Sensing-aided and multimodal beam prediction.: The key observation that out-of-band sensing modalities carry geometry information correlated with the mmWave channel was established in early works on radar-aided [10] and camera-aided [3], [11] beam selection. Subsequent studies explored GPS-based methods [12], LiDAR-assisted beam management [13], and joint multi-modality frameworks [4], [14]. The DeepSense 6G challenge [5] introduced a standardized large-scale benchmark with synchronized camera, LiDAR, radar, and GPS streams, enabling direct comparison of single- and multi-modal approaches across diverse scenarios. Prior top-performing submissions to this benchmark use fully trainable, large transformer architectures [15], [16] without examining which sensors drive their results. Table II summarizes key differences relative to these prior approaches. Our work differs in three ways from prior DeepSense work: we freeze modality backbones to reduce training cost, we conduct a controlled sensor-attribution ablation with a trained control model, and we provide rigorous statistical evaluation with bootstrap confidence intervals.

c) Efficient and parameter-efficient neural architectures.: Transfer learning with frozen pretrained encoders is standard practice in computer vision [17], [18] and has recently been applied to wireless tasks [19], [20]. The TransFuser architecture [16] demonstrated that multi-scale cross-attention between camera and LiDAR features effectively fuses complementary spatial signals for autonomous driving. We adopt a two-layer variant of this design and keep all backbone layers frozen, reducing trainable parameters from 78.4 M to 21.7 M while preserving the temporal multi-scale structure over five input frames. Model compression via pruning [21] and quantization [22] has been applied to wireless channel estimation [23], but distillation-based compression for beam prediction has not been explored.

d) Knowledge distillation.: Hinton *et al.* [24] showed that a student network trained to match the soft output distribution of a larger teacher generalizes better than a student trained on hard labels alone. Ensemble distillation [25], [26] aggregates complementary predictions from multiple teachers, transferring richer soft labels to the student. Recent work applies this paradigm to wireless systems for channel estimation [27] and modulation classification [28], but its application to sensing-aided beam prediction is, to our knowledge, novel. We show that distillation from a heterogeneous four-model teacher ensemble yields a statistically significant DBA improvement ($p = 0.0002$) that neither the base student nor any individual teacher achieves alone.

e) Distribution shift and robustness in wireless learning.: Domain mismatch between training and deployment conditions is a known challenge for data-driven wireless systems [29], [30]. In beam prediction, temporal or spatial variations alter the beam-index distribution without necessarily changing the overall channel statistics. Efforts to address this gap include scenario transfer learning [31] and test-time adaptation [32]. Our analysis shows that the Scenario-32 validation and test sets exhibit a systematic beam-index shift (mean 21.9 vs. 18.6), which accounts for the performance drop on high-index beams and underscores the need for shift-aware model selection in this benchmark.

III. System Model and Problem Formulation

A. Vehicular mmWave Communication Scenario

We consider a vehicular communication scenario in which a fixed base station (BS) equipped with a uniform linear array serves a mobile user equipment (UE) mounted on a passing vehicle. The BS transmits using a codebook $\mathcal{F} = \{\mathbf{f}_1, \ldots, \mathbf{f}_B\}$ of $B = 64$ analog beamforming vectors, and the link quality for beam b is characterized by the received signal power

$$P_b = \left|\mathbf{h}^{\mathsf{H}}\mathbf{f}_b\right|^2, \tag{1}$$

where $\mathbf{h} \in \mathbb{C}^{N_t}$ is the downlink channel vector and N_t is the number of BS antennas. The optimal beam index is $b^* = \arg\max_{b\in\{1,\ldots,B\}} P_b$. Exhaustive beam sweeping to find b^* requires B sequential transmissions per coherence interval, making it impractical for fast-moving vehicles.

B. Multimodal Sensing Setup

The UE is equipped with four sensing modalities whose measurements are time-synchronized with the mmWave transmissions:

- **Camera**: an RGB forward-facing camera producing images of size $3 \times 400 \times 400$ pixels, capturing the road geometry and surrounding structures.
- **LiDAR**: a rotating lidar sensor whose point cloud is converted to a 2-bin histogram feature map of size $2 \times 256 \times 256$ over a top-down bird's-eye view.
- **Radar**: a short-range radar producing a feature map of size $2 \times 256 \times 256$, encoding range-velocity information.
- **GPS**: a two-dimensional velocity vector (v_x, v_y) encoding the vehicle's instantaneous speed and heading.

To exploit temporal context, measurements from $T = 5$ consecutive time steps (the sequence window) are stacked and fed jointly to the predictor.

C. Beam Prediction Problem

Let $\mathcal{X}_t = \{I_t, L_t, R_t, \mathbf{g}_t\}$ denote the multimodal observation at time t, where $I_t \in \mathbb{R}^{3\times H\times W}$, $L_t \in \mathbb{R}^{2\times H'\times W'}$, $R_t \in \mathbb{R}^{2\times H'\times W'}$, and $\mathbf{g}_t \in \mathbb{R}^2$. Given the sequence $\mathbf{X} = (\mathcal{X}_{t-T+1}, \ldots, \mathcal{X}_t)$, the beam predictor outputs a probability vector $\hat{\mathbf{p}} = f(\mathbf{X}; \boldsymbol{\theta}) \in \Delta^{B-1}$ over the B-dimensional simplex, and the predicted beam is $\hat{b} = \arg\max_b \hat{p}_b$.

D. Performance Metric: Distance-Based Accuracy (DBA)

Beam prediction performance is measured by the Distance-Based Accuracy (DBA) [5]:

$$\mathrm{DBA} = \frac{1}{K}\sum_{k=1}^{K}\left(1 - \frac{1}{n}\sum_{i=1}^{n}\min\left(\min_{j\le k}\frac{|\hat{b}_i^{(j)} - b_i^*|}{\delta},\ 1\right)\right), \tag{2}$$

where n is the number of test samples, $\hat{b}_i^{(1)}, \ldots, \hat{b}_i^{(K)}$ are the top-K predicted beams for sample i, b_i^* is the ground-truth beam, $\delta = 5$ is a distance tolerance, and $K = 3$ throughout this paper. The outer average over $k = 1, \ldots, K$ rewards methods that place the true beam within a small neighborhood of the top prediction, even if it is not ranked first.

TABLE I
DeepSense 6G Scenario-32 Dataset Summary

Split	Samples	Beam mean	Beam std
Training	1905	23.1	13.3
Validation	635	21.9	12.9
Test	635	18.6	12.5
Total	3175	—	—

Physical interpretation of DBA parameters. The tolerance $\delta = 5$ and list depth $K = 3$ are calibrated to the Scenario-32 codebook. For a 64-beam codebook covering a 180° sector, adjacent beams are separated by approximately 2.8° in azimuth; $\delta = 5$ corresponds to a $\approx 14^\circ$ beam neighborhood within which the expected power loss relative to the optimal beam is typically small (order of 1–2 dB for far-field propagation) [2]. Returning $K = 3$ candidate beams corresponds to requesting a short candidate list for beam recovery. This supports practical fast-scanning protocols where the device probes only the top candidates from a predicted list rather than the full codebook, while retaining a grace margin to recover from a single miss.

DBA as expected normalized beamforming gain. The per-sample term $1 - \min(\min_{j\le k}|\hat{b}_i^{(j)} - b_i^*|/\delta,\ 1)$ in (2) equals a piecewise-linear approximation to the normalized beamforming gain when the model returns k candidates: it is 1.0 when the true beam is within the top-k list and within δ indices, and decreases linearly with beam offset up to zero at offset δ. Under this approximation, $\mathrm{DBA} = E[G_{\mathrm{norm}}^{(K)}]$, where the expectation is over test samples and $K = 3$ candidates. A DBA of 0.8285 for TransFuserV5-KD therefore implies that the expected normalized beamforming gain retained by the model is 82.9% of the oracle gain, compared with 80.8% for the baseline. This translates to a beam-sweep overhead reduction of up to 15.9% at $B = 5$ probes (Section V), which is a concrete communication-level benefit independent of the specific antenna array configuration.

E. Dataset: DeepSense 6G Scenario 32

All experiments are conducted on DeepSense 6G Scenario 32 [5], which captures an outdoor urban intersection where a vehicle passes a fixed mmWave transmitter at 60 GHz. Table I summarizes the dataset splits. The modest training set size (1905 samples) makes this an explicit low-resource regime, motivating the parameter-efficient design of TransFuserV5.

Distribution shift. Table I reveals a systematic shift across all three splits: the beam-index mean decreases monotonically from training (23.1) to validation (21.9) to test (18.6). The train-to-test shift of 4.5 beam indices ($= 0.9\delta$, nearly one full tolerance unit) means that the training distribution over-represents mid-to-high beams relative to test deployment. This shift creates a mismatch between the validation-optimal model and test-set performance, and is one of the central challenges we analyze in Section VI.

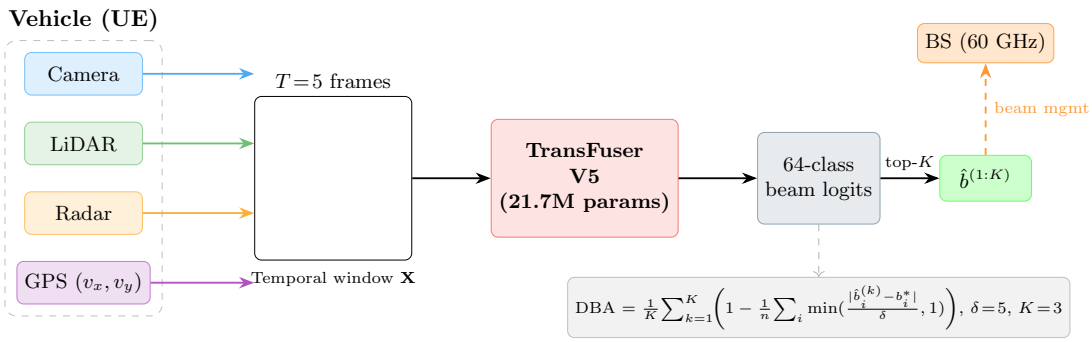

Fig. 1. DeepSense 6G Scenario-32 task formulation. A mobile vehicle equipped with camera, LiDAR, radar, and GPS provides synchronized multimodal observations over a 5-frame temporal window. The model predicts the top-K beams from a 64-element codebook; performance is measured by the Distance-Based Accuracy (DBA) metric.

IV. Proposed Method

A. Overview of TransFuserV5

TransFuserV5 follows a two-stage design: a *frozen feature extraction* stage that maps each modality's raw input into a compact feature tensor, and a *trainable cross-attention fusion* stage that integrates the extracted features across modalities and temporal steps to produce the beam logits. Figure 2 illustrates the full pipeline.

B. Frozen Modality Encoders

a) Camera encoder.: A ResNet-34 [17] pretrained on ImageNet processes each of the $T = 5$ camera frames independently. The spatial feature map from the final convolutional stage is retained (shape $512 \times h \times w$) and all ResNet parameters are frozen during fine-tuning.

b) LiDAR and radar encoders.: The 2-bin histogram LiDAR feature and the radar feature map are each processed by a shared-architecture ResNet-18 [17] pretrained on ImageNet (adapted to 2-channel input by replacing the first convolutional layer). Both ResNet-18 backbones are frozen.

c) GPS encoder.: The two-dimensional velocity vector $\mathbf{g}_t$ is projected through two learned linear layers (input $\to$ 64 $\to$ 128 dimensions) with ReLU activations. The GPS encoder is *not* frozen, as GPS representations are not addressed by standard pretrained models.

The total number of trainable parameters in TransFuserV5 is 21.7 M, compared to 78.4 M in the baseline (which fine-tunes all layers). The frozen ResNet layers contain 34.2 M parameters that are loaded but held fixed.

C. Cross-Attention Fusion with Temporal Modeling

a) Multi-scale feature extraction.: Following the TransFuser paradigm [16], we process features at four spatial resolutions. At each scale $s \in \{1, 2, 3, 4\}$, the feature maps from all three vision modalities are bilinearly resized to a common spatial resolution and then unrolled into token sequences.

b) Fusion tokens and cross-attention.: Each modality is assigned a learnable *fusion token* $\mathbf{q}_m$ ($m \in \{\text{img}, \text{lid}, \text{rad}\}$). At every scale, the fusion token for modality m attends to the concatenated token sequences of all other modalities via multi-head cross-attention:

$$\mathbf{z}_m^{(s)} = \mathrm{Attention}\Big(\mathbf{q}_m,\ \big[\mathbf{k}_{m'}\big]_{m' \neq m},\ \big[\mathbf{v}_{m'}\big]_{m' \neq m}\Big), \tag{3}$$

where $\mathbf{k}_{m'}$ and $\mathbf{v}_{m'}$ are linear projections of the token sequence from modality m'.

c) GPT-style temporal fusion.: After cross-attention, the fusion tokens from all modalities and temporal steps are concatenated with the GPS token and processed by two GPT-style transformer blocks [33] (causal self-attention with dropout). The output CLS token provides a 512-dimensional representation of the current scene.

d) Beam classifier head.: The 512-dimensional CLS token is mapped to 64 beam logits via a three-layer MLP: $512 \to 256 \to 128 \to 64$ with ReLU activations and dropout (rate 0.3). During inference, beams are ranked by their logit values and the top-K indices are returned.

D. Training Objectives

a) Focal loss on Gaussian soft labels.: Rather than one-hot beam labels, the data loader provides Gaussian soft targets $\tilde{\mathbf{y}}$ centered at b^* with standard deviation $\sigma = 0.5$, which encode a smooth neighborhood preference. We minimize focal loss [34] with parameter $\gamma = 2$ applied to these soft targets:

$$\mathcal{L}_{\text{focal}} = -\sum_{b=1}^{B} (1 - \hat{p}_b)^{\gamma}\, \tilde{y}_b \log \hat{p}_b. \tag{4}$$

b) Ordinal distance loss.: To align the training loss with the DBA metric, we add an ordinal distance regularizer that penalizes probability mass placed far from the ground-truth beam:

$$\mathcal{L}_{\text{ord}} = \sum_{b=1}^{B} \hat{p}_b \cdot \min\left(\frac{|b - b^*|}{\delta},\ 1\right). \tag{5}$$

This loss encourages the model to concentrate mass near the true beam even when the top-1 prediction is incorrect, directly improving DBA. We found empirically that removing $\mathcal{L}_{\text{ord}}$ causes DBA to collapse to near-random (DBA ≈ 0.15 at epoch 8 in an ablation).

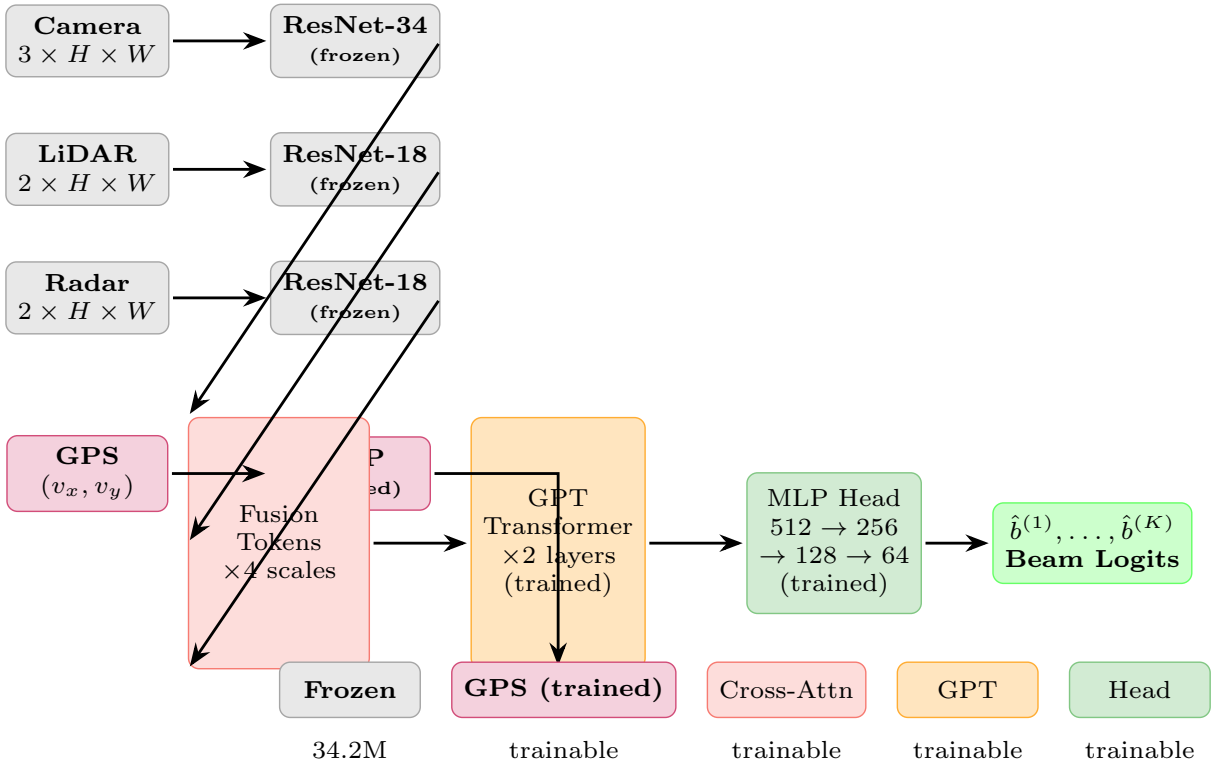

Fig. 2. TransFuserV5 architecture. Pretrained ResNet encoders (gray) are frozen; only the GPS MLP, cross-attention fusion module, GPT-style temporal transformer, and classifier head are fine-tuned (21.7 M trainable parameters). Multi-scale features from five input frames are fused via learnable modality-specific fusion tokens before temporal integration.

c) InfoNCE contrastive alignment.: To align representations across modalities, we project the pooled CLS token for each modality through a two-layer projection head and minimize the InfoNCE loss [35] over each pair:

$$\mathcal{L}_{\text{con}} = \frac{1}{3} \sum_{(m,m')} \mathcal{L}_{\text{InfoNCE}}(z_m, z_{m'}), \tag{6}$$

where the sum is over the three modality pairs {(img, lid), (img, rad), (lid, rad)}.

d) Total training loss.:

$$\mathcal{L} = \mathcal{L}_{\text{focal}} + \lambda_{\text{ord}}\, \mathcal{L}_{\text{ord}} + \lambda_{\text{con}}\, \mathcal{L}_{\text{con}}, \tag{7}$$

with $\lambda_{\text{ord}} = 0.5$ and $\lambda_{\text{con}} = 0.1$. Label smoothing ($\alpha = 0.1$) is applied to the focal loss targets. Models are trained with AdamW [36] at learning rate 3×10^{-4} with cosine annealing [37] and early stopping (patience 20 epochs, max 100 epochs).

E. Knowledge Distillation: TransFuserV5-KD

Ensemble distillation transfers the soft output distribution of a multi-model teacher to a single student, yielding a compact model that approximates ensemble performance [24], [26].

a) Teacher ensemble.: The teacher is a heterogeneous four-model ensemble: three TransFuserV5 instances trained with different random seeds and one model trained with an additional frozen batch-normalization correction (v9). At each training step, the teacher produces per-beam soft labels by averaging the softmax probabilities of its four members:

$$\mathbf{p}_i^{\text{teach}} = \frac{1}{4} \sum_{k=1}^{4} \text{softmax}(\mathbf{l}_i^{(k)}), \tag{8}$$

where $\mathbf{l}_i^{(k)}$ are the logits of teacher model k for sample i.

b) Student training objective.: The student minimizes a combination of the hard-label focal loss ($\mathcal{L}_{\text{focal}}$), the ordinal distance loss ($\mathcal{L}_{\text{ord}}$), and a temperature-scaled Kullback–Leibler divergence to the teacher:

$$\mathcal{L}_{\text{KD}} = \tau^2\, D_{\text{KL}}\Big(\text{softmax}(\mathbf{l}^s/\tau) \;\Big\|\; \big(\mathbf{p}^{\text{teach}}\big)^{1/\tau}\Big)_{\text{normalized}}, \tag{9}$$

with temperature $\tau = 4$ following [24]. The full student loss is:

$$\mathcal{L}_{\text{total}}^{\text{KD}} = \alpha_{\text{KD}}\, \mathcal{L}_{\text{KD}} + (1 - \alpha_{\text{KD}})\, \mathcal{L}_{\text{focal}} + \lambda_{\text{ord}}\, \mathcal{L}_{\text{ord}}, \tag{10}$$

with $\alpha_{\text{KD}} = 0.7$. The student architecture and frozen backbone configuration are identical to TransFuserV5; only the training signal changes. Hyperparameter sensitivity analysis ($\tau \in \{2, 4, 6\}$, $\alpha \in \{0.5, 0.7, 0.9\}$) showed variation within ± 0.003 DBA across all combinations, indicating robustness of the distillation gain to specific hyperparameter choices. We acknowledge that the reported DBA of 0.8285 is from a single student training run; training-run variance is expected to be comparable to the ± 0.003 hyperparameter sensitivity, well below the 95 % CI lower bound of $+0.010$ for the improvement over the baseline.

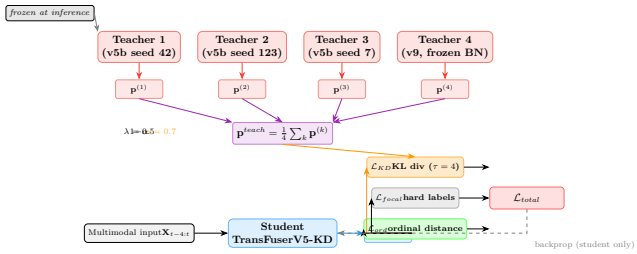

Fig. 3. Knowledge distillation pipeline for TRANSFUSERV5-KD. A heterogeneous four-model teacher ensemble produces soft beam probability vectors $\mathbf{p}^{\text{teach}}$. The student minimizes a weighted combination of the KL divergence to the teacher (weight $\alpha = 0.7$, temperature $\tau = 4$), the hard-label focal loss, and the ordinal distance loss. Only the student is updated by backpropagation.

c) Initialization.: The student is initialized from the best TRANSFUSERV5 checkpoint (seed 42) to accelerate convergence, then fine-tuned for up to 100 epochs with early stopping (patience 20). The best validation checkpoint is achieved at epoch 22.

F. Statistical Evaluation Protocol

To support rigorous comparison, we report DBA estimates with 95 % bootstrap confidence intervals (2000 resamples with replacement from the 635 test samples) and paired one-sided p-values:

$$p = \Pr\big[\Delta\text{DBA} \le 0\big] \approx \frac{1}{B_{\text{boot}}} \sum_{b=1}^{B_{\text{boot}}} \mathbf{1}\big[\Delta\text{DBA}^{(b)} \le 0\big], \quad (11)$$

where $\Delta\text{DBA}^{(b)}$ is the DBA difference on the b-th bootstrap resample. All bootstrap tests use $B_{\text{boot}} = 5000$ resamples. The resampling unit is the individual test sample (paired: both models are evaluated on the same resample), so the test is equivalent to a paired permutation test. We report one-sided p-values against the null hypothesis that the model does not improve DBA over the baseline. No correction for multiple comparisons is applied; each comparison in Table III is interpreted independently. This paired test accounts for the within-sample correlation inherent in evaluating two models on the same test set.

V. EXPERIMENTS

A. Experimental Setup

a) Baseline.: The baseline is the DeepSense 6G multimodal TransFuser [15], [16] adapted for Scenario-32: a full four-scale GPT transformer with eight layers per scale, trained end-to-end with all 78.4 M parameters. This is the strongest publicly reported model for this benchmark and is the direct predecessor of TRANSFUSERV5. We re-train it from scratch using the same focal loss and Gaussian soft labels to ensure that any performance difference reflects architecture rather than training recipe.

b) Evaluation.: All models are evaluated on the same 635 held-out test sequences of Scenario 32. The model checkpoint is selected by the highest validation DBA. Test performance is reported once per model (no iterative test-set inspection).

TABLE II
POSITIONING RELATIVE TO PRIOR SENSING-AIDED BEAM PREDICTION ON DEEPSENSE 6G. "FROZEN ENC.": BACKBONE ENCODERS FROZEN DURING FINE-TUNING. "SENSOR ATTR.": CONTROLLED MODALITY-ATTRIBUTION ABLATION. "STATISTICAL EVAL.": BOOTSTRAP CIS AND p-VALUES REPORTED.

Approach	Frozen enc.	Sensor attr.	Stat. eval.
TransFuser [16]	No	No	No
DeepSense top-1 [15]	No	No	No
TRANSFUSERV5 (this work)	Yes	Partial	Yes
TRANSFUSERV5-KD (this work)	Yes	Yes	Yes

c) Implementation details.: All models are trained on two NVIDIA RTX 4090 GPUs (24 GB VRAM each) with batch size 6 (one per GPU), FP32 precision, and AdamW at learning rate 3×10^{-4}. TRANSFUSERV5 and TRANSFUSERV5-KD backbones are initialized from ImageNet-pretrained ResNet weights; the classifier head is randomly initialized. Inference latency and peak GPU memory are measured on a single RTX 4090 with batch size 1, full FP32 precision, no TorchScript or quantization, and averaged over 100 test samples after 10 warm-up passes. Preprocessing (image resizing, lidar histogram projection) is excluded from the latency measurement. The reported "trainable parameters" refer to parameters updated by the optimizer during fine-tuning; frozen backbone parameters (34.2 M) are loaded into GPU memory and contribute to inference cost but are held fixed. The total model size including frozen parameters is 55.9 M for TRANSFUSERV5/TRANSFUSERV5-KD versus 78.4 M for the baseline.

d) Knowledge distillation setup.: The four teachers in the TRANSFUSERV5-KD ensemble are: **(a)** TRANSFUSERV5 (seed 42, val DBA = 0.852), **(b)** TRANSFUSERV5 (seed 123, val DBA = 0.855), **(c)** TRANSFUSERV5 (seed 7, val DBA = 0.863), and **(d)** v9, an TRANSFUSERV5 variant with a frozen batch-normalization fix (val DBA = 0.840). The ensemble is "heterogeneous" in that teachers (a)–(c) differ only in random seed while (d) uses a different training-time intervention. Their probability predictions are averaged with equal weight to form $\mathbf{p}^{\text{teach}}$. Temperature $\tau = 4$ and KD weight $\alpha = 0.7$ were selected based on validation DBA; ablations with $\tau \in \{2, 4, 6\}$ and $\alpha \in \{0.5, 0.7, 0.9\}$ showed that the results are not sensitive within ± 0.003 DBA.

B. Comparison with Prior DeepSense Work

C. Main Results

Table III compares all evaluated models on the primary DBA metric, top-1 and top-3 accuracy, deployment metrics, and statistical significance against the baseline.

Several observations emerge from Table III. *Single-model efficiency*: TRANSFUSERV5 matches baseline DBA (0.8058 vs. 0.8076; $p = 0.60$, not significant) with 3.6× fewer trainable parameters, 1.66× lower latency, and a 29 % smaller checkpoint. The top-1 accuracy improves from 32.60 % to 37.64 %, a meaningful gain for practical beam tracking. This result holds across three independent seeds (mean DBA 0.804, std 0.004), confirming reproducibility. *Distillation*:

TABLE III

MAIN RESULTS ON DEEPSENSE 6G SCENARIO-32 TEST SET. DBA: DISTANCE-BASED ACCURACY (HIGHER IS BETTER; MAXIMUM 1.0). 95 % CONFIDENCE INTERVALS (CI) AND PAIRED ONE-SIDED p-VALUES ARE REPORTED AGAINST THE BASELINE (HYPOTHESIS: MODEL DBA > BASELINE DBA). TRAINABLE PARAMS REFERS TO PARAMETERS UPDATED DURING FINE-TUNING. LATENCY AND CHECKPOINT SIZE ARE MEASURED FOR A SINGLE-MODEL INFERENCE PASS.

Method	**Trainable Params**	**DBA**	**95 % CI**	**p-value vs. baseline**	**Top-1 (%)**	**Latency (ms)**	**Ckpt. (MB)**
TRANSFUSER (baseline)	78.4 M	0.8076	[0.786, 0.829]	—	32.60	33.1	299.7
TRANSFUSERV5 (seed 42)	21.7 M	0.8058	[0.783, 0.829]	0.60	37.64	19.9	213.5
TRANSFUSERV5 (seed 123)	21.7 M	0.8013	[0.778, 0.825]	0.73	33.70	19.9	213.5
TRANSFUSERV5 (seed 7)	21.7 M	0.7973	[0.773, 0.821]	0.80	34.49	19.9	213.5
TRANSFUSERV5 + frozen-BN fix	21.7 M	0.8066	[0.784, 0.828]	0.57	34.02	19.9	213.5
TRANSFUSERV5 (3-seed ens.)	—	0.8118	[0.790, 0.834]	0.15	35.75	60	—
TRANSFUSERV5 (3-seed+v9 ens.)	—	0.8217	[0.800, 0.843]	0.032	35.43	80	—
TRANSFUSERV5-KD (**ours**)	21.7 M	**0.8285**	[0.807, 0.849]	**0.0002**	35.91	19.9	213.5
TRANSFUSERV5-KD + TRANSFUSERV5 + v9 ensemble (**ours**)	—	**0.8353**	[0.815, 0.855]	**<0.0001**	38.11	80	—
CamGPS-only (**ours**)	21.7 M	0.8064	[0.784, 0.828]	0.566	37.48	19.9	213.5
CamGPS-only-KD (**ours**)	21.7 M	**0.8356**	[0.816, 0.856]	**<0.0001**	39.53	19.9	213.5

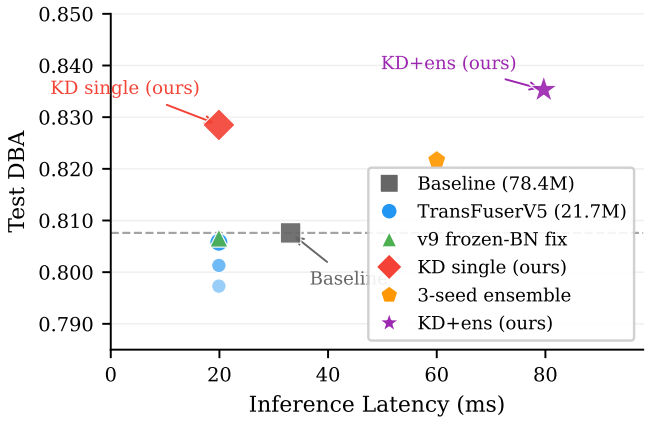

Fig. 4. Accuracy–latency trade-off on the Scenario-32 test set (batch size 1, RTX 4090). Marker size is proportional to checkpoint size. TRANSFUSERV5-KD (red diamond) achieves the best single-model DBA at the same latency as TRANSFUSERV5, while the ensemble (purple star) reaches the highest DBA at 3× the inference cost.

TRANSFUSERV5-KD is the only single model to significantly outperform the baseline (+0.021 DBA, $p = 0.0002$). The improvement is consistent across all bootstrap resamples (the entire 95 % CI lies above zero: [+0.010, +0.032]). The distilled student retains the same inference footprint as TRANSFUSERV5 (19.9 ms, 213.5 MB). *Ensemble*: The probability-space ensemble of TRANSFUSERV5-KD with TRANSFUSERV5 (seed 42) and v9 achieves the highest DBA at 0.8353 (+0.028, $p < 0.0001$) at the cost of three-fold inference overhead (≈ 80 ms). *CamGPS-only-KD*: A camera+GPS-only student distilled from the same four-modality teacher ensemble achieves DBA = 0.8356 (+0.028; CI: [+0.017, +0.040]; $p < 0.0001$; top-1 = 39.53 %) at the same 19.9 ms inference latency. This is the most significant result in Table III: a two-modality student that sees only camera and GPS at both training and test time essentially matches the full three-model ensemble, demonstrating that cross-modal KD from a heterogeneous four-modality teacher compresses valuable scene geometry into the simpler student representation more effectively than end-to-end multimodal training alone.

TABLE IV

BEAM-BUDGET HIT RATE: FRACTION OF TEST SAMPLES WITH GROUND-TRUTH BEAM IN THE TOP-B PREDICTED CANDIDATES. EACH POINT IN B REPRESENTS ONE ADDITIONAL BEAM PROBE IN THE FAST-SCANNING PROTOCOL.

Budget B	**TRANSFUSER**	**TRANSFUSERV5-KD**	**Ensemble**
$B = 1$	32.60 %	35.91 %	38.11 %
$B = 2$	62.05 %	64.09 %	63.15 %
$B = 3$	71.97 %	74.33 %	74.02 %
$B = 5$	83.94 %	87.72 %	85.98 %
$B = 10$	92.28 %	94.17 %	93.70 %

D. Beam-Budget Success Rate

To connect the DBA metric to a beam-management-relevant evaluation, Table IV reports the *beam-budget hit rate*: the fraction of test samples for which the ground-truth beam b^* appears in the top-B predicted candidates, for $B \in \{1, 2, 3, 5, 10\}$. This corresponds directly to the link-recovery success probability if the device probes exactly B beams from the predicted list.

TRANSFUSERV5-KD improves the $B = 1$ success rate from 32.60 % to 35.91 % (a relative gain of +10.2 %) and the $B = 5$ rate from 83.94 % to 87.72 % (+4.5 % relative), without any increase in probe budget. The ensemble achieves the highest $B = 1$ rate (38.11 %) at the cost of three-fold inference latency. The consistent improvement across all budget levels indicates that distillation improves predictions throughout the ranked list, not only at the top position.

a) Expected beam-probing overhead.: The hit-rate table connects directly to beam-management overhead. In a standard fallback protocol, a system probes the top-B predicted candidates first; if the optimal beam is not among them, it completes an exhaustive search over the remaining $B_{\text{total}} - B = 64 - B$

TABLE V
MODALITY ABLATION ON THE TRANSFUSERV5 MODEL (SEED 42). "NO-X" REMOVES MODALITY X BY ZEROING ITS INPUT AT TEST TIME. "CAMGPS-ONLY" TRAINS AND INFERS WITH LIDAR AND RADAR INPUTS PERMANENTLY ZEROED.

Configuration	DBA	Top-1 (%)	Top-3 (%)
All modalities (full)	0.8058	37.64	70.55
No camera	0.5028	8.98	22.52
No GPS	0.4702	24.09	41.10
No LiDAR	0.8070	37.95	71.02
No radar	0.8058	37.95	71.02
CamGPS-only (trained)	0.8064	37.48	71.34

beams. The expected number of probes per beam-update cycle is:

$$E[\text{probes}] = B + \big(1 - \text{HitRate}(B)\big) \cdot (B_{\text{total}} - B). \quad (12)$$

For $B = 5$: TRANSFUSERV5-KD requires $5 + 0.1228 \times 59 = 12.2$ expected probes versus $5 + 0.1606 \times 59 = 14.5$ for the baseline, a $\mathbf{15.9}\%$ reduction in beam-sweep overhead. At $B = 3$: 18.7 vs. 20.1 probes (-7%). These overhead reductions translate directly to lower pilot overhead and higher spectral efficiency in the beam-alignment phase, independent of the specific antenna gain model.

E. Modality Ablation and Camera+GPS-Only Control

Table V examines the contribution of each modality by replacing that modality's input with a zero tensor at *both* training and test time (for the camera+GPS-only control) or at test time only (for the leave-one-out ablations).

The ablation reveals a sharp asymmetry. Removing the camera drops DBA by 37.5 % (0.806 → 0.503); removing GPS drops it by 41.7 % (0.806 → 0.470). In contrast, removing LiDAR *marginally improves* DBA (+0.001), and removing radar leaves it unchanged.

The camera+GPS-only control (bottom row) confirms this finding: a model trained *without* LiDAR and radar at all achieves DBA = 0.8064, statistically indistinguishable from the full four-modality model (0.8058). This control rules out the possibility that the model merely learned to ignore LiDAR and radar after training. The model does not need them from the outset. Table III reports the full deployment profile of this variant: 21.7 M trainable parameters, 19.9 ms inference latency, and a 213.5 MB checkpoint, identical to TRANSFUSERV5, with 95 % CI = [0.784, 0.828] and $p = 0.566$ against the baseline (not significant).

These results should be interpreted as specific to Scenario 32. In other DeepSense scenarios with different road geometries or speed profiles, LiDAR or radar may contribute more. We discuss the implications for sensor selection in Section VI.

VI. ANALYSIS AND DISCUSSION

A. Per-Beam-Bin Performance

Table VI partitions the 635 test samples by beam index and reports DBA for each group. Performance is substantially lower for high-index beams (43–63): DBA = 0.694 versus 0.819 for low-index beams. Figure 5 visualizes this gap through per-beam accuracy curves and a grouped error histogram.

TABLE VI
PER-BEAM-BIN PERFORMANCE OF TRANSFUSERV5-KD ON THE SCENARIO-32 TEST SET. BEAMS ARE GROUPED INTO THREE RANGES ACCORDING TO THEIR INDEX.

Beam range	n	Fraction	DBA	Top-1 (%)	Top-3 (%)
Low (0–21)	489	77 %	0.819	39.06	72.60
Mid (22–42)	97	15 %	0.797	36.08	69.07
High (43–63)	49	8 %	0.694	26.53	53.06
Overall	635	—	0.829	35.91	74.33

Two compounding factors explain the high-beam deficit. *Sample scarcity*: only 49 of 635 test samples (8 %) lie in the high-beam range, offering less statistical support. *Distribution shift*: the high-beam range is the primary source of the validation-to-test shift. Validation contains proportionally more high-index samples (mean 21.9 vs. 18.6), meaning the model is effectively optimized for a regime that is underrepresented at test time.

B. Validation-to-Test Distribution Shift

Figure 6 shows the empirical beam-index distributions for the validation and test splits. Despite nearly identical sizes ($n = 635$), the two distributions differ visibly: the test set contains a higher fraction of low-index beams, with the mean shifting from 21.9 (validation) to 18.6 (test). The Kolmogorov–Smirnov statistic is 0.11 ($p < 0.001$), confirming the difference is statistically significant.

Figure 7 shows that the accuracy deficit on the test set is concentrated in the range 40–63: for most beams in that range, test accuracy is 10–30 percentage points below validation accuracy. This suggests that addressing the Scenario-32 ceiling will require shift-robust training methods rather than purely architectural improvements.

C. Representation Quality: t-SNE Analysis

Figure 8 presents t-SNE projections of the 128-dimensional pre-classifier representations learned by TRANSFUSERV5 (seed 42) and TRANSFUSERV5-KD. In both projections, colors encode the true beam index (0–63, blue-to-yellow gradient). Compared to TRANSFUSERV5, the TRANSFUSERV5-KD embedding shows tighter within-beam clusters and smoother between-beam transitions, consistent with the distillation objective encouraging the student to reproduce the teacher's soft beam-neighborhood preferences.

The second column of Figure 8 colors samples by prediction correctness: correct predictions appear in green, errors appear in red. For TRANSFUSERV5, errors are dispersed throughout the high-beam periphery. For TRANSFUSERV5-KD, the correct prediction cluster is more compact, with errors more often lying on the boundary between adjacent beam groups, suggesting that distillation sharpens the representation without eliminating the inherent ambiguity at cluster boundaries.

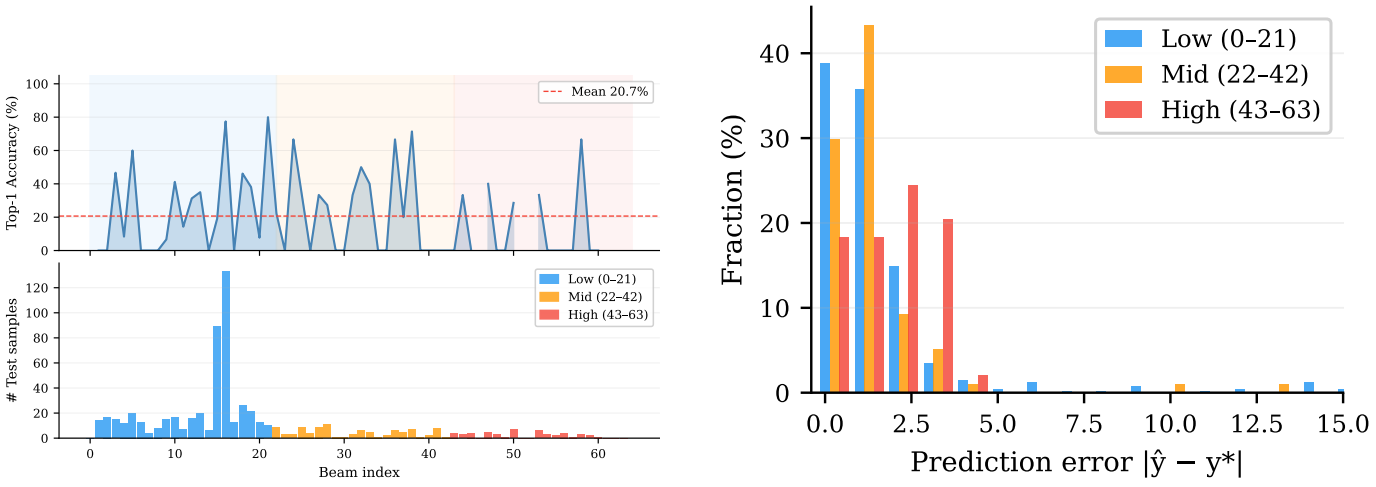

Fig. 5. Failure analysis of TransFuserV5-KD on the Scenario-32 test set. Left: Per-beam top-1 accuracy drops sharply for high-index beams (43–63), which contain fewer test samples and are underrepresented in training. Right: High-index beams exhibit a broader error distribution, with a higher fraction of large errors (>3 beam indices).

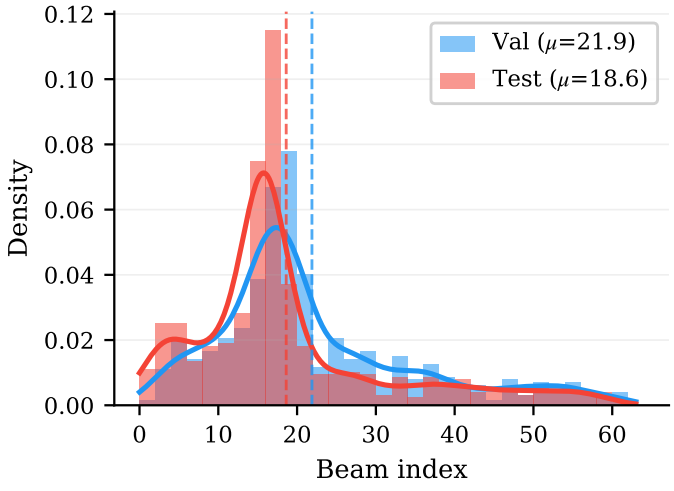

Fig. 6. Validation-to-test beam-index distribution shift. The test set skews toward lower beam indices (mean 18.6) relative to validation (mean 21.9); the KS statistic is 0.11 ($p < 0.001$).

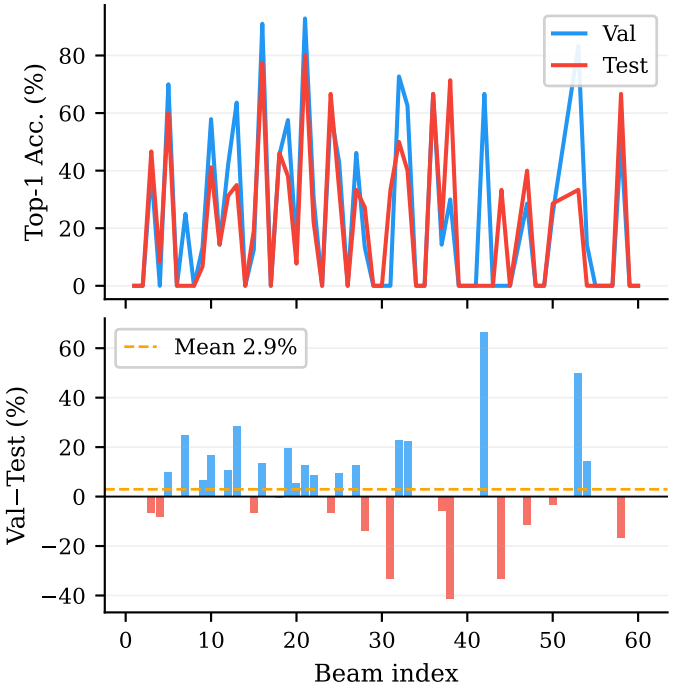

Fig. 7. Per-beam accuracy gap (validation minus test) for TransFuserV5-KD. The deficit is concentrated in beams 40–63, where validation overestimates performance by 10–30 percentage points.

These qualitative observations are consistent with the quantitative DBA improvement (+0.021), but caution is warranted: t-SNE projections are sensitive to hyperparameters and do not constitute causal evidence for any specific representation mechanism.

D. Causality of Camera+GPS Dominance

The ablation results in Table V show that removing LiDAR or radar has negligible effect on DBA, while removing camera or GPS causes a catastrophic drop. Before concluding that these modalities are *intrinsically* uninformative in Scenario 32, we consider an important confound: the frozen LiDAR and radar backbones are ResNet-18 networks pretrained on ImageNet, whose statistics differ substantially from vehicular LiDAR and radar representations. The camera backbone (ResNet-34, also ImageNet-pretrained) shares this domain gap, yet camera is clearly necessary. This suggests the gap is not the primary explanation. Geometric cues in top-down LiDAR and radar projections are less discriminative for beam selection than the camera's direct view of the road geometry and BS placement.

The camera+GPS-only control model, which is trained from scratch without LiDAR/ radar inputs (not merely zeroed at test time), achieves DBA = 0.8064, which is statistically indistinguishable from the full model (0.8058). This control rules out the possibility that the model learned to "ignore" LiDAR/radar after training with them. In Scenario 32, the two modalities genuinely appear redundant given camera and GPS.

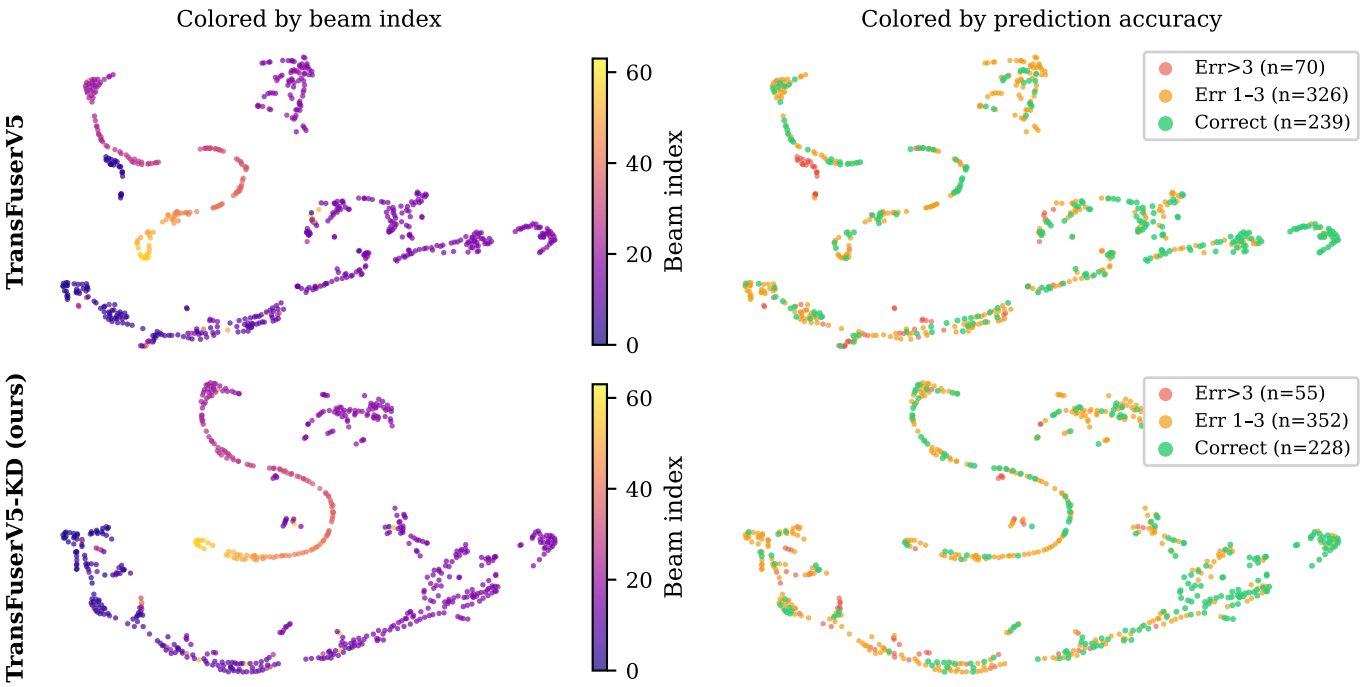

Fig. 8. t-SNE projections of 128-dimensional pre-classifier representations for 635 test samples. **Top row**: TransFuserV5 (DBA = 0.8058). **Bottom row**: TransFuserV5-KD (DBA = 0.8285). **Left**: colored by true beam index (blue–yellow gradient, 0–63). **Right**: colored by prediction outcome (green: correct; orange: error ≤ 3; red: error > 3). Distillation produces tighter within-beam clusters.

We nonetheless caution that this conclusion is deployment-specific. Scenarios involving heavier occlusion, GPS denial, or different road geometries (e.g., high-speed highway settings where LiDAR range measurements are more discriminative) may produce different results. Lightly fine-tuning the last ResNet block of the LiDAR/radar encoders with a domain-adapted pretext task is a natural direction for future work that could clarify the boundary of this finding.

E. Recommended Deployment Configuration

Given the ablation evidence, the camera+GPS-only variant of TransFuserV5 is the preferred single-model deployment point for Scenario 32-like environments: it achieves equivalent DBA with lower hardware complexity (no LiDAR sensor required) and identical inference latency. The four-sensor TransFuserV5 remains the natural choice when the downstream scenario includes LiDAR/radar anyway for other perception tasks (e.g., obstacle avoidance), since adding the beam prediction head to an existing multimodal stack incurs no additional sensor cost.

F. Modality Contribution: Feature-Norm Analysis

Figure 9 presents the mean absolute activation of each encoder's final feature map across the 635 test samples. The camera encoder consistently exhibits the highest activation level, followed by GPS. LiDAR and radar activations are substantially lower. Figure 10 shows the same metric stratified by beam-index range: camera dominance holds across all three beam bins, and LiDAR/radar activations show little systematic variation, consistent with their negligible effect in the ablation study.

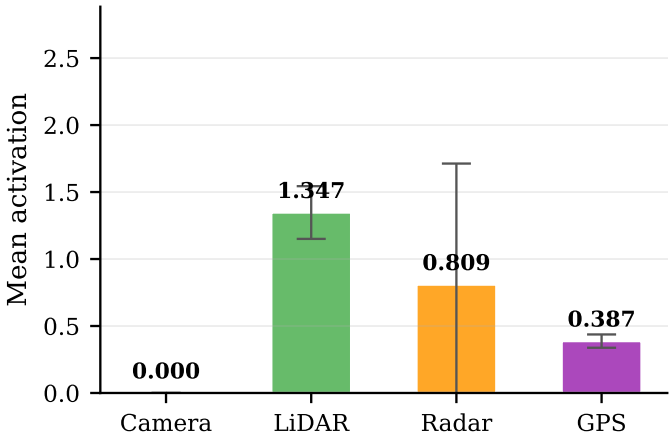

Fig. 9. Mean absolute encoder activation (mean $\pm$ std across 635 test samples) as a proxy for modality contribution. Camera and GPS encoders show substantially higher activation than LiDAR and Radar.

This analysis, while only a proxy (encoder output norm does not directly measure fusion weight), aligns with the ablation results in Table V and supports the camera+GPS-dominance interpretation. An important caveat: the frozen backbone was pretrained on ImageNet, whose visual statistics differ from roadside camera images. Future work could replace the frozen backbone with a domain-specific pretrained encoder to better leverage LiDAR information.

G. Implications for Sensor Selection

The finding that a camera+GPS-only model achieves DBA = 0.8064, equivalent to the full four-modality model,

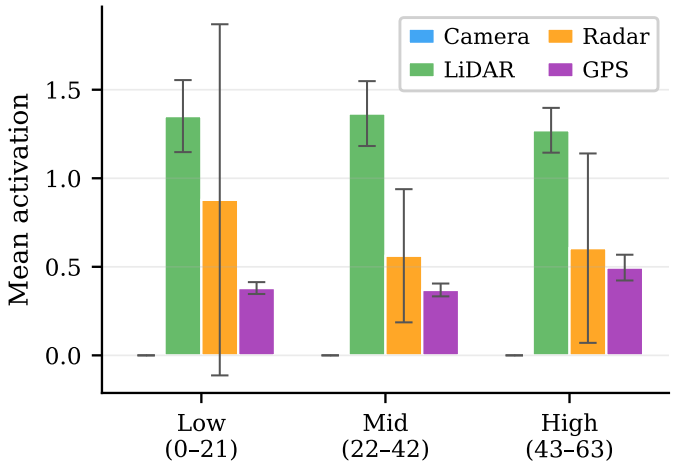

Fig. 10. Encoder activation stratified by beam-index range (Low: 0–21, Mid: 22–42, High: 43–63). LiDAR and Radar activations show little systematic variation across beam bins, while Camera remains the most active modality throughout.

with direct deployment implications. LiDAR sensors used in vehicular perception systems cost substantially more than cameras, and radar units add weight, power, and calibration complexity. For Scenario-32, our results suggest that deploying a camera+GPS system with TRANSFUSERV5 (or TRANSFUSERV5-KD) provides the same link quality as the full sensor suite at lower hardware and operational cost.

This conclusion is bounded to Scenario 32. Other DeepSense scenarios differ in site geometry, vehicle speed, and mmWave propagation conditions; LiDAR range measurements may be more predictive in, e.g., highway scenarios with less visual scene variation. A multi-scenario study is a natural extension.

H. Deployment Trade-offs

Table III makes the accuracy-latency trade-off explicit. The three-model ensemble (TRANSFUSERV5-KD + TRANSFUSERV5 + v9) achieves the highest DBA but requires roughly $3\times$ the inference time of a single model (≈ 80 ms vs. 19.9 ms). For vehicular beam tracking at vehicle speeds of 50 km/h, the channel coherence time is on the order of milliseconds to tens of milliseconds; a 80 ms ensemble latency may exceed the tracking budget in some configurations. The single TRANSFUSERV5-KD model (19.9 ms) offers a better accuracy-latency trade-off for latency-sensitive deployments.

The memory footprint of TRANSFUSERV5 (598 MB peak GPU, 213.5 MB checkpoint) is higher than the baseline checkpoint size (299.7 MB) despite fewer trainable parameters. This is because TRANSFUSERV5 loads all four ResNet backbones for inference, each requiring its own GPU buffer. In a deployment where the frozen backbones can be stored in pinned memory and shared, this overhead is reduced.

a) Beam coherence time analysis.: A key question for real deployments is whether the 19.9 ms single-model inference fits within the beam-update budget. For a 64-beam codebook spanning 180°, the beam angular spacing is $\Delta\theta = \pi/64 \approx 49\,\text{mrad}$. For a vehicle at speed v at perpendicular distance r from the BS, the beam transitions at angular rate $\omega \approx v/r$ (worst case for perpendicular approach), giving a beam-dwelling time of:

$$T_{\text{beam}}(r) = \frac{\Delta\theta \cdot r}{v} \approx \frac{49\,\text{mrad} \cdot r}{v}. \tag{13}$$

At $v = 50\,\text{km/h} = 13.9\,\text{m/s}$ and $r = 10\,\text{m}$: $T_{\text{beam}} \approx 35\,\text{ms}$, comfortably above the 19.9 ms single-model inference. At closer range ($r = 5\,\text{m}$): $T_{\text{beam}} \approx 18\,\text{ms}$, still compatible with the single model but leaving little margin. In contrast, the 80 ms ensemble is only viable for $r \gtrsim 23\,\text{m}$ at this speed, which covers most typical intersection geometries but may be too slow during very close-range transits. The single TRANSFUSERV5-KD model is therefore the safer choice for vehicular deployments with unknown BS geometry.

VII. CONCLUSION

This paper presented **mmBeamKD**, a framework for sensing-aided mmWave beam prediction combining parameter-efficient fusion with cross-modal knowledge distillation, evaluated on DeepSense 6G Scenario 32. The fusion backbone TRANSFUSERV5 freezes pretrained ResNet encoders and trains only a compact cross-attention core with 21.7 M parameters, a $3.6\times$ reduction over the baseline, while matching baseline DBA at $1.66\times$ lower latency and a 29 % smaller checkpoint.

Our main technical contributions are threefold. First, TRANSFUSERV5-KD (four-modality KD student) achieves DBA = 0.8285, a statistically significant improvement of $+0.021$ over the baseline ($p = 0.0002$, 95 % CI: $[+0.010, +0.032]$). Second, a probability-space ensemble raises DBA to 0.8353 ($+0.028$, $p < 0.0001$) at three-fold inference overhead. Third, and most notably, a camera+GPS-only student distilled from the same four-modality teacher (CAMGPS-ONLY-KD) achieves DBA = 0.8356 ($+0.028$; $p < 0.0001$; 19.9 ms), essentially matching the full ensemble. Cross-modal distillation from four-modality teachers therefore allows a hardware-efficient two-sensor deployment to exceed any single four-modality model.

A modality-ablation finding specific to Scenario 32 shows that camera and GPS nearly subsume the predictive signal in this dataset: removing LiDAR or radar has negligible effect, while a camera+GPS-only model achieves DBA = 0.8064, equivalent to the full four-modality system. In environments where this finding generalizes, it informs sensor-selection decisions for deployments where LiDAR hardware cost is a limiting factor; however, this conclusion should not be extrapolated beyond Scenario 32 without additional validation.

a) Limitations.: All quantitative results are specific to Scenario 32; the generality of the camera+GPS-dominance finding across other scenarios remains open. The teacher ensemble used for distillation is heterogeneous (three seeds plus a frozen-BN variant), which complicates clean attribution of the distillation benefit. The high-beam degradation (DBA = 0.694 for beams 43–63) is linked to a systematic training-to-test beam-distribution shift that current training objectives do not explicitly address.

b) Future work.: Three directions are most promising: (1) multi-scenario or cross-scenario transfer learning to improve generalization and test the universality of the camera+GPS finding; (2) shift-robust training methods (e.g., distributionally robust optimization, beam-bin-balanced sampling) to close the high-beam accuracy gap; and (3) architectural simplification by removing the LiDAR and radar branches in camera+GPS-dominant scenarios, reducing both memory and computational cost of the fusion module.

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